Origins and Evolution of Language
Week 6 tutorial
Tutor notes

Comments for tutors are in italics.

The main aim for today is to read and talk about a very interesting paper with results I personally find fascinating and potentially slightly challenging for my world view, as discussed below!

This week's lecture reading was about vocal learning, and in the lecture I focussed on grammar learning. The workshop picks up on both topics, and looks at sequence-learning biases in vocal learners.

For this week's workshop, read James & Sakata (2017) and Tchernichovski & Lipkind (2017), which is a helpful one-page summary/comment on the James & Sakata paper. You could also watch this brief video explainer of the experiment: …

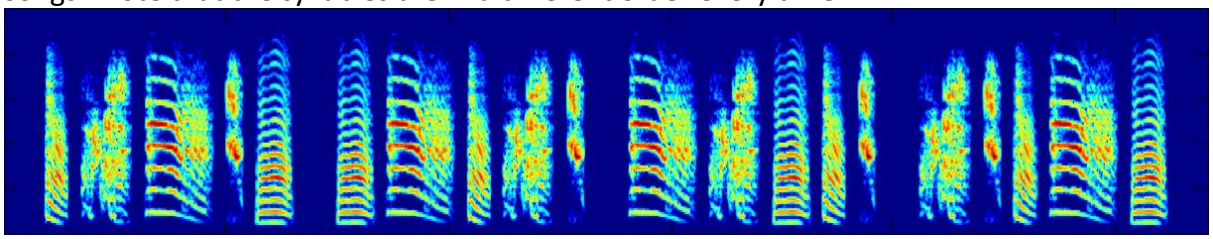
James & Sakata is fairly technical; don't worry about the technicalities of the statistical analysis(unless you are in to that kind of thing), but try to get the gist of what they are describing; the Tchernichovski & Lipkind comment is quite helpful for this too.

It might be worth pointing out that knowing what details you can probably gloss over and which ones are important for your specific purposes is one of the skills you have to pick up when reading papers! If you are in hyper-critical mode and focussed on low-level details you might care about exactly how they conducted these analyses, but in the questions I set here the focus is more on the general findings and the relationship to human language, in which case I can take the details of the analysis on trust. It is definitely not the case that you need to understand every detail of every paper you read.

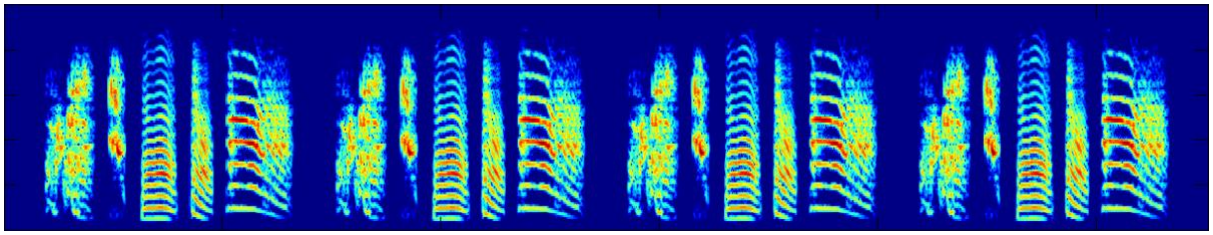
To get a feel for natural "wild-type" zebra finch song, have a listen to the songs below - 6 wild-type song examples of naturally-tutored birds: Wild-type song 1, Wild-type song 2, Wild-type song 3, Wild-type song 4, Wild-type song 5, Wild-type song 6. You could compare these to the songs produced by sol-called "isolate" birds who are raised without any song input, and produce songs that are distinctively different (to the expert ear!): they are a bit longer, they have more repetition of individual syllables rather than repeating the entire syllable sequence, and are generally a bit more screechy. Isolate song A, Isolate song B, Isolate song C.

These are courtesy of Olga Fehér. She tells me zebra finch song is beautiful, I don't hear it, which is a constant source of incomprehension and disappointment to her – to me it sounds like a clown car crashing into a wall. Anyway, I am including the isolate song as another data point suggesting that zebra finches have a fairly strong template for what their song should look like – there are differences between normally-tutored and isolate song, but they are less extreme than you would expect to see with humans in similar circumstances!

To get a sense of the unstructured training data the experimentally-tutored birds from James & Sakata were trained on, click on this spectrogram to listen to 4 examples of 5-syllable songs - note that the syllables are in a different order every time.



And here's a synthesised sample of 4 repetitions of a song using 5 syllables in a consistent order.



And here are two example songs sung by their experimentally-tutored birds: the common b-e-a-d song, and a less typical d-b-a-e-a song.

You might have to talk them through the spectrograms in conjunction with the audio.

For the second question below, you need some example song syllables to reconstruct what kind of songs the experimentally-produced birds might sing - here are your 5 syllables.

- Syllable a (short duration, lower pitch)
- Syllable b (medium duration, higher pitch)
- Syllable c (medium duration, lower pitch)
- Syllable d (longest duration, medium pitch)
- Syllable e (shortest duration, highest pitch)

Some of their findings relate to tendencies in the song in how they order syllables of different duration / pitch, as well as syllable-specific tendencies, which is why I provide (very rough!) duration/pitch info.

Questions:
- What is the James & Sakata paper about? What did they do? What did they find?

They are interested in biases in the way zebra finches learn and produce song sequences, and whether these biases can explain common patterns in birdsong (and potentially human speech and music). Normally-reared zebra finches show consistent patterns in how they arrange song syllables. To investigate whether these patterns arise simply from copying what they hear, James & Sakata tutored young zebra finches with songs made up of five syllable types arranged in randomized sequences, so there was no structure in their input – they heard every possible combination of all 5 notes. Despite this, the birds produced songs with similar structures: particular syllables tended to occur at the beginning and end, shorter and higher-frequency syllables tended to occur in the middle, and pitch tended to alternate between adjacent syllables. This suggests that biases in learning (and possibly constraints on vocal production) shape the songs birds produce, rather than their songs being determined

only by structure in their input. This suggests that are constraints on the form of zebra finch song that exert themselves during learning, driving the emergence of more natural, more wild-type song structure even in birds who were exposed to randomly-structured input – potentially a kind of universal grammar for zebra finches!

- One of their results relates to how syllable duration and frequency (pitch) influence the position of syllables in songs (Figures 2C and 2D in the paper) - syllables in the middle tend to differ systematically on these features from syllables at the beginning or end. Based on these results, and given the information above on duration and pitch, where would you expect syllables a-e might be most likely to appear? You might not be able to confidently guess for all syllables. Does this correspond to what they show in Figure 2B in terms of the positional preferences of these syllable-types?

Based on the figure it looks like middle syllables tend to be shorter and higher-pitched than syllables at the beginning or end of songs – I think the birds can in principle change the duration/pitch of a note so that it "fits" better where they want it, but as far as I can make out that also influences their preferred placement of the syllables. Based on that: syllable e really likes to be in the middle of songs (short, high-pitch), and should be preferred at edges (long, lower pitch). The other ones are less clear where these preferences re. duration and pitch would push them – b and c are perhaps better suited to the edges since they are long (but b is higher-pitched), a is short so should be preferred in middle positions (but is not high-pitched).

This hopefully leads into the final part of the question: it seems to me that the positional preferences of the notes, as shown in the "skew" rating in figure 2B, are not entirely predictable from duration and pitch. There are other pitch-related constraints going on (e.g. the preference to alternate pitch), and as mentioned above the birds can also change the duration/pitch of a note to adjust it based on where it occurs, but even so Figure 2B implies some additional more arbitrary biases for where they like to put syllables.

- Based on their findings, which of the following syllable sequences do you thing would be most likely to be sung by one of their birds, and why? e-d-b-e, c-a-b-e-d, c-e-a-d.

e-d-b-e should be really dispreferred – it's got short high-pitched syllables at the periphery and long in the middle, plus it violates the positional preferences of e (likes to be in the middle), d (likes to be at the end) and b (likes to be at the start).

c-a-b-e-d is less bad in terms of duration and individual syllable placement – it's got a b in the middle which violates b's preferences, but otherwise there's nothing terrible there. But it doesn't alternate pitches, which the birds otherwise like.

c-e-a-d should be the best of these from a zebra-finch perspective – it should be fine in terms of duration and individual syllable placement, and it also alternates pitch.

- What aspect of their findings strikes you as most interesting or surprising from the perspective of someone interested in human language? Or if you think these results aren't relevant to understanding human language, why not?

For me the most striking and challenging aspect of their findings is that they are probably quite convincing evidence for an innate template that shapes how song develops in zebra finches, and the obvious implication is that human language works in the same way. The work on iterated learning we talked about in week 1 and return to in week 8-9 instead emphasises the interaction between learning and communication in shaping human language, and in general I think of learning constraints as being relatively weak and quite general (e.g. a preference for simpler grammars over more complex ones, rather than a highly detailed template for what grammars should look like), which may not to be how it works for zebra finches. Of course zebra finches might be radically different from humans, but since this is the species that has been studied most closely we have to take note of it.

- If you can imagine running similar experiments with humans, what do you think would happen? How different would the methods have to be to show interesting results? This question is intended to be about the factors you think are crucial in understanding human language transmission, rather than the ethical issues!

Obviously this kind of thing couldn't be done with humans from birth for multiple ethical and practical reasons (although it perhaps has been done in the past, see https://en.wikipedia.org/wiki/Language_deprivation_experiments), so try not to get too sidetracked on that! But there are several interesting and relevant data points we can draw on.

My colleague here in Edinburgh, Prof. Jenny Culbertson, also runs experiments where she exposes human learners to artificial languages that are similar to the James & Sakata input in some respects – for example, involving noisy/probabilistic ordering, or even lacking data on the relative order of elements – and then tests those participants on their ordering preferences and looks for regularities in the orders they produce. This work is mostly conducted with adults or older kids (age 5+) who obviously have additional extensive experience with natural languages, but she finds evidence for quite strong ordering preferences in humans, which might parallel the James & Sakata results. However, my understanding of her interpretation of those results is that they reflect either quite general simplicity preferences (e.g. use the same order across phrase types) or preferences for how semantic relations are linearised in linguistic form (semantic distance should roughly be reflected in sequence distance), rather than being entirely arbitrary learning biases. Maybe the same sort of argument could be made for the James & Sakata data.

In weeks 8-9 we'll also return to the iterated learning paradigms we talked about already in week 1, where we present participants with a structureless artificial language (a bit like the James & Sakata input) and show how linguistic structure (arbitrary symbols, duality of patterning, compositionality) gradually emerges – over multiple episodes of learning and transmission, not in a single step , and also communication plays a crucial role, but these experiments again suggest a role for learning biases in shaping language learning in humansHowever, in these cases our experimental results lead me to believe the learning bias is relatively weak (unlike in zebra finches) and many of the constraints on what the linguistic system looks like come from communicative constraints. This is where I find the results of this paper challenging –work on zebra finches is perhaps the best model we currently have, and it

shows that innate learning biases play a crucial role, so I am sort of committed to a view that while these results are very interesting, I feel like the important mechanisms shaping linguistic universals in humans are probably rather different!

In week 9 we'll talk about homsigners, deaf children of hearing parents, who are exposed to gestural input which is deficient in several respects (not like perfectly-randomised experimental input in this paper of course, but lacking the structural properties of sign language) and have been argued to introduce some important structural devices that may be independent of the surrounding spoken language. That might be evidence for quite strong learning biases in human learners. We'll talk about that in a few weeks.